Algorithm Prime Triad: Candidate Intake, Adapter Descent, and Continuation Control

Jeremy H. Carroll

Version 1.0.0 — 2026-09-26

Contents

Why a Researcher Should Care . . . 1
Refusal Box & Scope Fence . . . 2
Mathematical Foundations & Notation . . . 2
1. Prime -1 & Prime 000: Candidate Proposal Engine & Local Declaration Order . . . 3
1.1 CandidatePacket & IntakeTrace Schema (OE001-01) . . . 3
1.2 Prime 000 Local Packet Declaration Order (OE001-02) . . . 4
1.3 Layered Multi-Carrier Lenses & LensSpan Model (OE001-03) . . . 4
1.4 Scoped Terminalization (OE001-04) . . . 4
2. Prime 001: Adapter Descent Soundness & Transport Verification . . . 5
2.1 Dual Prime 001 Placement Protocol (OE002-01) . . . 5
2.2 Adapter Descent Soundness Rule (OE002-02) . . . 5
2.3 Refusal & Debt Taxonomy (OE002-03) . . . 5
3. Prime 002: Master Continuation Predicate & Anti-Cheat Control . . . 5
3.1 Master Control Loop & Predicate $\mathsf{CanContinue}_n$ (OE003-01) . . . 5
3.2 Six Anti-Cheat Halting Laws (OE003-02) . . . 6
3.3 Bounded Operational Termination (OE003-03) . . . 6
3.4 Local Queue Scheduler (OE003-04) & CODA Taxonomy (OE003-05) . . . 6
4. CODA Recursive Re-Entry and C^n Ascent Protocol . . . 7
4.1 CODA Recursive Re-Entry Protocol (OE004-01) . . . 7
4.2 Lawful C^n Ascent Protocol (OE004-02) . . . 8
4.3 Stop Conditions: GTLA Stop vs. CODA-A Global Residual (OE004-03) . . . 8
4.4 Observer Envelopes & Evaluator Defect Linearity (OE004-04, OE004-05, OE004-06) 8
5. Summary & Claim-ID Index . . . 9
6. References . . . 10

Series: 008_engine_runtime / 000_oe_prime_algorithms
Series position: OE-000 (Part I — Algorithm Prime Triad Master Monograph)
Author: Jeremy H. Carroll
Date: 2026-09-26
Version: 1.0.0 (Unified Monograph combining Prime -1, 000, 001, 002, and CODA C^n Ascent)
Document class: Chapter / stem paper

Why a Researcher Should Care

Every recursive evaluator faces a fatal failure mode: it can always add a dimension, rename an unresolved residual, or re-enter an evaluation locus, thereby appearing to make progress forever without discharging anything.

The **Algorithm Prime Triad (000, 001, 002)** establishes the foundational operational control law of the Omega Engine:

1. **Prime -1 & 000:** Enforces strict locus grounding and a 7-step local packet declaration order (Grounding → Carrier → Observer → Primitive Difference → ZeroRel → Witness → Quotient).
2. **Prime 001:** Governs intake and inter-carrier transport soundness, enforcing that translation into an `OmegaPacket` does not confer validity and tracking all lost data in explicit loss ledgers (L_{adapter}).
3. **Prime 002:** Enforces the master continuation control law ($\mathsf{CanContinue}_n$) and six anti-cheat halting laws, guaranteeing finite, budgeted stopping.
4. **CODA C^n Ascent:** Formalizes recursive re-entry $\Omega_{000}(L)$ at any locus without minting false terminal authority.

Without a strict declaration, transport, and halting calculus, agentic execution swarms degenerate into endless prompt retries, un-grounded carrier hallucinations, and authority laundering.

Refusal Box & Scope Fence

```
No automatic conversion of raw prompts into valid cochains.
No degree, lift, or authority laundering.
No un-bounded recursive evaluation.
No un-grounded carrier hallucinations.

Candidate proposals under Prime -1 carry status: proposed_only
and confer no validity until Prime 000 admission.
Step 10 terminalization emits LOCAL_GREEN_FOR_SCOPE,
never an un-scoped global green certificate.
```

Mathematical Foundations & Notation

The operational calculus operates over discrete finite simplicial complexes $K = (V, E, F)$ equipped with exact rational arithmetic over $\mathbb{Q}$:

1. **Chains and Cochains:** C_0, C_1, C_2 denote the rational vector spaces spanned by vertices, oriented edges, and oriented faces. Cochain spaces $C^k = \text{Hom}(C_k, \mathbb{Q})$ represent evaluation functions.
2. **Coboundary Operators:** The discrete derivative $d_0 : C^0 \to C^1$ assigns to potential ϕ the edge difference $(d_0\phi)(u, v) = \phi(v) - \phi(u)$. The curl operator $d_1 : C^1 \to C^2$ evaluates circulation around faces $(d_1 h)(u, v, w) = h(u, v) + h(v, w) - h(u, w)$, satisfying $d_1 \circ d_0 = 0$.
3. **Exact Quotient Potential (Φ_1):** Given a 1-cochain $h \in C^1$, the quotient potential is the exact minimum L1 distance to the subspace of coboundaries (gauge transforms):

$$\Phi_1(h) := \min_{\phi \in C^0} \|h - d_0\phi\|_1$$

   This is evaluated exactly over $\mathbb{Q}$ via rational linear programming, yielding a dual witness vector $z \in [-1, 1]^{|E|}$ with $d_0^* z = 0$ certifying the zero duality gap: $\Phi_1(h) = \langle z, h \rangle$.

4. **Closure Charge (Γ_2):** The 2-obstruction measuring non-closure is:

$$\Gamma_2(h) := \|d_1 h\|_1$$

5. **Discrete Hodge Decomposition:** For an inner product with positive edge weights W, any 1-cochain decomposes orthogonally over $\mathbb{Q}$ into exact, harmonic, and coexact components:

$$h = d_0\phi + h_{\text{harm}} + d_1^*\psi$$

where $h_{\text{harm}} \in \ker(d_1) \cap \ker(d_0^* W)$.

---

1. Prime -1 & Prime 000: Candidate Proposal Engine & Local Declaration Order

Prime -1: Candidate Proposal $\longrightarrow$ Prime 001 Intake Check
$\longrightarrow$ Prime 000 Validation $\longrightarrow$ DDA Execution

Requiring an end-user to manually invent carriers, primitive differences, or observers forces them to perform internal mathematical tasks. **Prime -1 (Carrier Proposal Engine / Packet Compiler)** acts as the front-end intake compiler:

> The Omega Engine does not begin by knowing the carrier.
> It begins by proposing candidate carriers under scoped admission.

1.1 CandidatePacket & IntakeTrace Schema (`OE001-01`)

Prime -1 ingests a raw artifact and emits candidate `CandidatePacket` structures. Pre-admission refusals live strictly in an `IntakeTrace` (`enters_dda_trace: false`) to prevent contaminating the operational `DdaTrace`: `[DEFINITION]` (`OE001-01`)

```yaml
IntakeTrace:
  raw_artifact_hash: "doc_001_sha256"
  candidate_locus: "lines 1-12"
  candidate_packet:
    artifact_id: "doc_001_ref"
    locus_id: "locus_frontmatter_01"
    problem_family: "claim | proof | narrative | code"
    primitive_difference_proposal: "metadata mismatch"
    carrier_proposal: "document metadata block"
    zero_rel_proposal: "canonical metadata equality"
    observer_proposal: "frontmatter schema checker"
    witness_available: true
    quotient_or_refusal: "E or M"
    adapter_confidence: "high"
    status: "proposed_only"
  intake_adapter_status: "ADMITTED | REFUSED_UNTYPED"
  enters_dda_trace: false
```

Prime 000 receives each candidate and emits one of seven formal responses: `ACCEPT_DECLARATION` · `REFUSE_MISSING_CARRIER` · `REFUSE_MISSING_OBSERVER` · `REFUSE_MISSING_WITNESS` · `REFUSE_MISSING_QUOTIENT` · `SPLIT_PACKET` · `REQUEST_LIFT`.

1.2 Prime 000 Local Packet Declaration Order (OE001-02)

Every admitted packet MUST be declared according to the 7-step local runtime declaration order: [DEFINITION] (OE001-02)

```
0. Grounding (cryptographic raw text / AST / commit hash anchor)
1. Carrier (simplicial complex / AST graph / discrete carrier)
2. Observer packet / detector (resolution, probe, gauge)
3. Primitive difference (cochain h in C1 or defect functional)
4. ZeroRel rule (gauge equivalence / coboundary relation)
5. Witness or refusal route (replay certificate chi or typed refusal)
6. Optional quotient, degree, lift, or terminal policy
```

1.3 Layered Multi-Carrier Lenses & LensSpan Model (OE001-03)

A complex artifact (paper, codebase, financial plan) is **not** a single monolithic cochain complex. It is a **bundle of carriers** extracted via layered partial spans: [SCOPE FENCE] (OE001-03)

$$A \xleftarrow{s_L} S_L \xrightarrow{p_L} K_L$$

where S_L is the selected source span, K_L is the extracted carrier, and span composition proceeds by pullback in the category $\mathbf{LensSpan}(A)$ under canonical inclusion subcomplexes:

$$\begin{aligned} L_0 \text{ (lexical)} &\longrightarrow L_1 \text{ (syntactic)} \longrightarrow L_2 \text{ (semantic)} \\ &\longrightarrow L_3 \text{ (proof)} \longrightarrow L_4 \text{ (rhetoric)} \longrightarrow L_5 \text{ (strategic)} \end{aligned}$$

1.4 Scoped Terminalization (OE001-04)

Terminalization (T) emits scoped local status rather than un-scoped global green: [OPERATIONAL RULE] (OE001-04)

$$\text{Step 10: } T \longrightarrow \text{Terminalize local repair pass: } \mathsf{LOCAL_GREEN_FOR_SCOPE}(\text{scope}, \text{remaining_obligations})$$

Terminalization evaluates the active cochain state and routes through two distinct terminal paths:

1. **Zero-Residual Terminalization ($T_{\mathbf{res}}$):** If the quotient residual vanishes ($\Phi_1 = 0$) and the closure charge is exact ($\Gamma_2 = 0$) under an adequate detector, the pass seals with LOCAL_GREEN_FOR_SCOPE. The emitted receipt registers the exact potential and witnesses, certifying completion strictly within the declared carrier scope.
2. **Terminal Failure / Refusal ($F_{\mathbf{term}}$):** If an open residual ($\Phi_1 > 0$) or closure failure ($\Gamma_2 > 0$) persists, terminalization is refused. The evaluator halts with typed terminal failure F_{term} (REFUSED_OPEN_RESIDUAL or REFUSED_BLIND_OBSERVER), preventing un-repaired defects from being sealed as green.

Under no circumstances does terminalization convert a local repair into an un-scoped global green certificate.

---

2. Prime 001: Adapter Descent Soundness & Transport Verification

2.1 Dual Prime 001 Placement Protocol (OE002-01)

To prevent un-grounded or invalid translations from contaminating the execution log, Prime 001 is deployed at **two distinct boundary points**: [OPERATIONAL RULE] (OE002-01)

$$\begin{aligned}\text{Prime (-1)} &\longrightarrow \text{Prime 001 [Intake Adapter]} \longrightarrow \text{Prime 000 [Admit]} \\ &\longrightarrow \text{DDA Runner} \longrightarrow \text{Prime 001 [Transport Adapter]} \longrightarrow \text{Prime 002 [Control]}\end{aligned}$$

1. **Intake Adapter Check:** Evaluates raw artifact → candidate packet translation before Prime 000 admission.
2. **Transport Adapter Check:** Evaluates inter-carrier transport steps whenever a DDA operation changes carrier.

2.2 Adapter Descent Soundness Rule (OE002-02)

Translation of a native claim into an OmegaPacket is a representation mapping, not a proof of validity. An intake or transport adapter $\alpha : C_{\text{source}} \to C_{\text{target}}$ is sound if and only if it preserves operations, equations, witnesses, and authority fences while recording all lost data in an explicit loss ledger L_{adapter}: [DEFINITION] (OE002-02)

```
AdapterContract:
  source_carrier: C_source
  target_carrier: C_target
  preserved_operations: [d0, d1]
  preserved_equations: [d1_d0_zero]
  preserved_witnesses: [dual_certificate]
  loss_ledger: L_adapter
```

If $L_{\text{adapter}} \neq \varnothing$, the adapter emits ADAPTER_DEBT and halts automatic promotion.

2.3 Refusal & Debt Taxonomy (OE002-03)

1. REFUSED_UNTYPED: Raw input cannot be mapped to any known carrier signature or observer probe.
2. ADAPTER_INADEQUATE: Candidate carrier mapping exists but fails boundary operator or witness preservation.
3. REFUSED_ADAPTER_DEBT: Transport across carrier boundaries failed adapter soundness checks (C · M · T).

---

3. Prime 002: Master Continuation Predicate & Anti-Cheat Control

3.1 Master Control Loop & Predicate $\mathsf{CanContinue}_n$ (OE003-01)

At stage n, continuation to stage $n+1$ is lawful if and only if predicate $\mathsf{CanContinue}_n$ evaluates to True: [DEFINITION] (OE003-01)

$$\begin{aligned}\mathsf{CanContinue}_n :\iff\ & \mathsf{OutstandingObligation}(r_n) \wedge \mathsf{HigherObligationWitness}(r_n) \wedge \\ & \mathsf{NextCarrierDeclared}(r_n) \wedge \mathsf{AdapterSound}_{n\to n+1} \wedge \\ & \mathsf{DetectorAdequate}_{n+1} \wedge \mathsf{MinimalLicensedLift}_{n\to n+1} \wedge \\ & \mathsf{SkeletonPreserved}_{n\to n+1} \wedge \mathsf{BudgetAvailable}_n \wedge \\ & \neg\mathsf{GTLAContinuationRefusal}_n\end{aligned}$$

The three-condition form groups these conjuncts into structural validity, resource availability, and absence of terminal refusal. The Lean theorem `Prime.cc9_implies_cc3` formalizes this equivalent regrouping.

3.2 Six Anti-Cheat Halting Laws (OE003-02)

Halting Law Constraints:

1. **No Degree Laundering:** Adding C^{n+1} requires an exhibited $(n+1)$-dimensional obligation.
2. **No Lift Laundering:** A representation extension must prove no same-carrier repair exists.
3. **No Anomalon Evasion:** A stable residual cannot be perpetually renamed to avoid terminalization.
4. **Old-Skeleton Preservation:** A continuation must preserve certified data or emit a loss ledger.
5. **Detector Revalidation:** Repacketization must revalidate detector adequacy at degree $n+1$.
6. **First-Unresolved-Degree Honesty:** If n_0 is unresolved, claims above n_0 are conditional: $\mathsf{CoherentUpTo}(n_0 - 1)$ and $\mathsf{ResidualAt}(n_0)$.

3.3 Bounded Operational Termination (OE003-03)

Given a finite non-replenished budget $B_{\text{tot}} \in \mathbb{N}$ (or scaled rational budget) where every counted operation debits the shared budget by at least a strictly positive cost floor $\delta_{\text{control}} > 0$ and no operation can begin without sufficient remaining budget ($B \geq \delta_{\text{control}}$), the number of executed operations is bounded by: `[CONDITIONAL THEOREM]` (OE003-03)

$$\#\text{ops} \leq \left\lfloor \frac{B_{\text{tot}}}{\delta_{\text{control}}} \right\rfloor$$

This bound (`Prime.ops_le_floor_div`) formalizes the natural-number arithmetic consequence of the non-replenished debit invariant ($n \cdot \delta \leq B$). Total runtime termination additionally assumes that each individual operational checker halts in finite time. If budgets are tracked in exact rationals $\mathbb{Q}$, the bound applies via scaling by the least common denominator.

3.4 Local Queue Scheduler (OE003-04) & CODA Taxonomy (OE003-05)

For local execution on a fixed carrier, Prime 002 acts as a finite residual queue scheduler. To prevent anomalon evasion (Law 3), progress requires strict lexicographical descent on the diagnostic tuple: `[OPERATIONAL RULE]` (OE003-04)

$$\text{Descent} : \Longleftrightarrow \ (\Gamma_{2,n+1}, \Phi_{1,n+1}, \text{defect_count}_{n+1}) <_{\text{lex}} (\Gamma_{2,n}, \Phi_{1,n}, \text{defect_count}_n)$$

Witnessed exploration or residual fingerprint changes represent exploratory transitions, not mathematical progress. Across steps on a fixed carrier, continuation requires quantitative potential descent $\Delta\Phi_1 \leq 0$ evaluated using the exact quotient norm $\|[h]\|_{Q1}$. Under a licensed lift across carriers ($C^n \to C^{n+1}$), Φ_1 may temporarily increase if the representation extension exposes higher-dimensional boundary obligations, provided the lift license explicitly certifies that no same-carrier descent is possible: `[FINITE THEOREM]` (`OE003-06`).

CODA Pipeline-Order Taxonomy (`OE003-05`) CODA formalizes local obstruction analysis as a strict 7-stage pipeline sequence, with Q as The Continuum: `[DEFINITION]` (`OE003-05`)

$$\begin{aligned} G \text{ (Grounding)} &\longrightarrow C \text{ (Carrier)} \longrightarrow J \text{ (Observer)} \\ &\longrightarrow \Delta \text{ (Primitive Difference)} \longrightarrow Z \text{ (ZeroRel)} \\ &\longrightarrow \chi \text{ (Witness)} \longrightarrow Q \text{ (Quotient)} \end{aligned}$$

1. **Grounding (G):** Raw artifact cryptographic digest and origin anchor.
2. **Carrier (C):** Explicit simplicial complex or discrete chain module representing the substrate.
3. **Observer (J):** Finite projection and resolution window specifying measurement capacity.
4. **Primitive Difference (Δ):** Concrete difference cochain $h \in C^1$ between competing models or steps.
5. **ZeroRel (Z):** Equivalence relation specifying exact triviality or gauge equivalence ($\operatorname{im} d_0$).
6. **Witness (χ):** Explicit dual witness certificate bounding the duality gap or confirming exact repair.
7. **Quotient (Q):** The Continuum, the quotient space $C^1/\operatorname{im}(d_0)$ hosting the residual obstruction.

Output status taxonomy:

- `LOCAL_GREEN_FOR_SCOPE`: Exact zero residual ($\Phi_1 = 0, \Gamma_2 = 0$) under an adequate observer.
- `REFUSED_UNTYPED`: Candidate locus lacking a declared carrier or grounding anchor.
- `ADAPTER_INADEQUATE`: Loss ledger identifies loss of essential invariants or unpreservable equations.
- `HALTED_BUDGET_EXHAUSTED`: Operational budget depleted ($B < \delta_{\text{control}}$).
- `HALTED_STAGNANT_PROGRESS`: Lookback window detects lack of strictly decreasing residual mass.
- `GTLA_STOP`: Terminal local stop halted with an unresolved obstruction that cannot be discharged under available resources.
- `GTTA_GLOBAL_RESIDUAL`: Ambient structural obstruction surviving into all admissible representations.

---

4. CODA Recursive Re-Entry and C^n Ascent Protocol

4.1 CODA Recursive Re-Entry Protocol (`OE004-01`)

Algorithm Prime 000 may be recursively re-entered at any CODA locus $L \in$ CODA (object, morphism, witness, residual, quotient, adapter, or terminal claim).

A **locus** L is the entity under examination, whereas a **carrier** C_L is the declared local mathematical structure in which comparisons and primitive differences on L are represented. In general, $L \neq C_L$ (identifying them without an explicit typing witness silently promotes an arbitrary locus into a carrier, bypassing the Prime -1 proposal engine and Prime 000 declaration order).

The recursive re-entry pipeline requires proposing and declaring a candidate carrier C_L for locus L before running Prime 000 on (L, C_L): `[PROTOCOL SPECIFICATION]` (`OE004-01`)

$$\boxed{\text{locus } L \longrightarrow \text{propose/declare carrier } C_L \longrightarrow \text{run Prime 000 on } (L, C_L)}$$

$$\Omega_{000}(L) \Longrightarrow (\text{Carrier} = C_L,\ \text{ZeroRel}_L,\ \text{Observer} = J_L,\ \text{Witness} = \chi_L,\ \text{Residual} = R_L,\ \text{Quotient} = Q_L)$$

subject to standard Prime 000 admission requirements.

4.2 Lawful C^n Ascent Protocol (`OE004-02`)

Degree ascent $C^k \longrightarrow C^{k+1}$ is permitted if and only if the paper declares: `[DEFINITION]` (`OE004-02`)

1. Carrier complex C^k, Differential $d^k : C^k \to C^{k+1}$, Next Differential $d^{k+1} : C^{k+1} \to C^{k+2}$, and exact nilpotency $d^{k+1} \circ d^k = 0$.
2. **Monotone Phase Embedding Definition:** Given an assumed monotone map embedding admission into execution, if admission precedes measurement ($\text{admit}(P) \leq \text{measure}(P)$), then the order-preserving structure guarantees that admission precedence reflects in execution for any packet pair where the map is monotone (`Prime.embedding_order_preserving`). This formalization proves the property of the assumed monotone structure; constructing a concrete causal trace embedding and differential nilpotency ($d \circ d = 0$) are separate, independent verification obligations.

4.3 Stop Conditions: GTLA Stop vs. CODA-A Global Residual (`OE004-03`)

- `GTLA_STOP:` Local authority construction halted due to missing carrier, detector, witness, quotient, or exhausted resources.
- `GTTA_GLOBAL_RESIDUAL:` Ambient structural obstruction R_U surviving transport into all admissible adapters.

4.4 Observer Envelopes & Evaluator Defect Linearity (`OE004-04`, `OE004-05`, `OE004-06`)

Observer Envelope Comparability (`OE004-04`) Observer envelopes J_1, J_2 are comparable only under shared grounding and explicit coordinate transformation bounds: `[SCOPE FENCE]` (`OE004-04`). Comparing evaluation scores across ungrounded or incomparable observers is strictly refused.

Evaluator Defect Linearity (`OE004-05`) Let Ω be a finite probability space with joint probability distribution $\mathbb{P}$. Let $E_g : A \otimes B \to [0,1]$ be a global composite evaluator and $E_1 : A \to [0,1], E_2 : B \to [0,1]$ be local component evaluators. Define the global pass rate and joint local pass rate: `[FINITE THEOREM]` (`OE004-05`)

$$\text{GPR} := \mathbb{E}[E_g(A \otimes B)], \qquad \text{LPR} := \mathbb{E}[E_1(A)E_2(B)]$$

The pointwise evaluator defect is defined as:

$$\Delta_E(A, B) := E_g(A \otimes B) - E_1(A)E_2(B)$$

By linearity of expectation over the joint probability law:

$$\text{GPR} - \text{LPR} = \mathbb{E}[E_g(A \otimes B) - E_1(A)E_2(B)] = \mathbb{E}[\Delta_E]$$

When evaluators decouple without cross-module obstruction ($\Delta_E = 0$ pointwise), local pass rates predict global pass rates identically (GPR = LPR). Note that zero expected defect ($\mathbb{E}[\Delta_E] = 0$) can also arise via cancellation unless $\Delta_E \geq 0$; only a non-zero expected defect forces a non-zero defect on some supported instances.

Six-Role EPK Overlay Diagnostic Scope Fence (`OE004-06`) The six-role EPK overlay enforces strict role boundaries: `[SCOPE FENCE]` (`OE004-06`). Evaluators, improvers, adversaries, certifiers, archivers, and governors evaluate strictly within their declared diagnostic scopes.

Engine self-examination ($\Omega_{000}(L_{\text{self}})$) and its anti-cheat queue upgrade rules are specified in `docs/residual_aware_can_continue.md`: `[MODEL CONSEQUENCE]` (`OE004-07`, `OE004-08`).

---

5. Summary & Claim-ID Index

Claim ID	Type	Section	Summary
`OE001-01`	`Definition`	§1.1	CandidatePacket & IntakeTrace schema definition
`OE001-02`	`Definition`	§1.2	Prime 000 7-step local packet declaration order
`OE001-03`	`Scope`	§1.3	Multi-carrier lens span pullback model
`OE001-04`	`Model consequence`	§1.4	Scoped terminalization `LOCAL_GREEN_FOR_SCOPE`
`OE002-01`	`Scope`	§2.1	Dual Prime 001 boundary placement (Intake & Transport)
`OE002-02`	`Definition`	§2.2	Adapter contract definition & loss ledger structure
`OE002-03`	`Definition`	§2.3	Refusal & debt taxonomy (`REFUSED_UNTYPED`, `ADAPTER_INADEQUATE`)
`OE003-01`	`Definition`	§3.1	Continuation predicate definition & equivalent 3-form regrouping

Claim ID	Type	Section	Summary
OE003-02	Protocol specification	§3.2	Six anti-cheat halting laws
OE003-03	Conditional theorem	§3.3	Bounded operational termination law
OE003-04	Operational rule	§3.4	Local queue scheduler & strict quotient descent condition
OE003-05	Definition	§3.4	CODA taxonomy in pipeline order with Q as The Continuum
OE003-06	Finite theorem	docs/residual_aware_can_continue.md	Residual-aware $\mathsf{CanContinue}_n$ with quantitative $\Delta\Phi \leq 0$ bound
OE004-01	Scope	§4.1	CODA recursive re-entry protocol $\Omega_{000}(L)$ over declared carrier C_L ($L \neq C_L$)
OE004-02	Definition	§4.2	Lawful ascent protocol & assumed monotone phase embedding
OE004-03	Definition	§4.3	GTLA_STOP vs. GTTA_GLOBAL_RESIDUAL stop conditions
OE004-04	Scope	§4.4	Observer envelope comparability
OE004-05	Finite theorem	§4.4	Evaluator defect linearity $\mathrm{GPR} - \mathrm{LPR} = \mathbb{E}[\Delta_E]$
OE004-06	Scope	§4.4	Six-role EPK overlay diagnostic scope fence
OE004-07	Model consequence	docs/residual_aware_can_continue.md	Self-examination re-entry path $\Omega_{000}(L_{\mathrm{self}})$ for engine state
OE004-08	Model consequence	docs/residual_aware_can_continue.md	Anti-cheat laws for self-proposal & progress queue upgrade

6. References

1. J. H. Carroll. *Discrete Deformation Calculus: A Finite Trace Calculus for Obstructions, Exact Repair, and Bounded Replay*, v1.0.0 (2026).

2. J. H. Carroll. *Local Green, Global Red I: Evaluated Path Kernels for Composed Agent Reliability*, v1.2.3 (2026). Zenodo, DOI: 10.5281/zenodo.22881911.
3. J. H. Carroll. *CODA-FOC: Finite Obstruction Calculus — Exact Quotient Diagnostics, Dual Witnesses and Two-Stage Coherence*. Omega Engine Research Programme (2026).
4. J. H. Carroll. *CODA-GTLA: General Theory of Layered Authority — A Leftover Class is Not Standing*. Omega Engine Research Programme (2026).
5. The Lean 4 Theorem Prover, v4.34.0. `https://lean-lang.org`.